Step 1 — Import libraries

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import
LinearRegression
```

Step 2 — Create a simple Salary DataFrame

```
data = {
    "Experience": [1, 2, 3, 4, 5, 6,
7, 8, 9, 10],
    "Salary": [25000, 28000, 32000,
35000, 40000,
               45000, 50000, 55000,
60000, 65000]
}

df = pd.DataFrame(data)

print(df)
```

You will get:

```
   Experience  Salary
0           1   25000
1           2   28000
2           3   32000
3           4   35000
4           5   40000
```

```
5             6   45000
6             7   50000
7             8   55000
8             9   60000
9            10   65000
```

Step 3 — Separate X and y

```
X = df[["Experience"]]
y = df["Salary"]
```

Important:

- `X` = Experience
- `y` = Salary

We use double brackets for `X` because `LinearRegression()` expects a 2D input.

Step 4 — Create the Linear Regression model

```
model = LinearRegression()
```

Step 5 — Train the model

```
model.fit(X, y)
```

Now the model has learned the relationship between **Experience** and **Salary**.

Step 6 — Find slope and intercept

```
print("Slope:", model.coef_[0])
print("Intercept:", model.intercept_)
```

The equation is:

Salary=mX+c

where:

- `m` = slope/coefficient
- `c` = intercept

You can print the complete equation:

```
print("Salary =", model.coef_[0], "*
Experience +", model.intercept_)
```

Step 7 — Make predictions

Suppose someone has **7 years of experience**:

```
new_data = [[7]]

prediction = model.predict(new_data)

print("Predicted Salary:",
prediction[0])
```

Step 8 — Get predictions for the entire DataFrame

```
df["Predicted_Salary"] =
model.predict(X)
```

```
print(df)
```

Now your DataFrame will have:

```
   Experience  Salary
Predicted_Salary
0           1   25000                ...
1           2   28000                ...
2           3   32000                ...
...
```

Step 9 — Plot the original data

First, let's see the actual salary points:

```
plt.scatter(X, y)

plt.xlabel("Experience (Years)")
plt.ylabel("Salary")
plt.title("Experience vs Salary")

plt.show()
```

Step 10 — Plot the Linear Regression line

This is the important part:

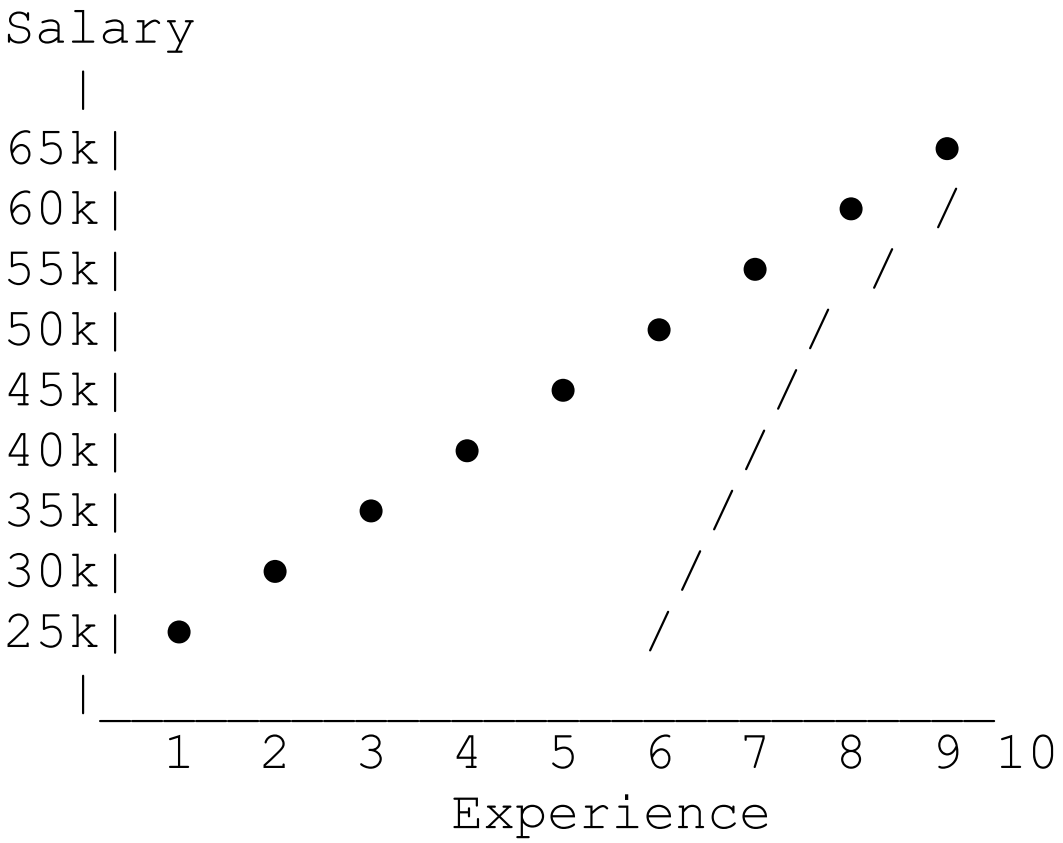
```
plt.scatter(X, y, label="Actual
Salary")
```

```
plt.plot(X, model.predict(X),
label="Regression Line")

plt.xlabel("Experience (Years)")
plt.ylabel("Salary")
plt.title("Linear Regression:
Experience vs Salary")

plt.legend()
plt.show()
```

You should see something like:



Complete code

Once you understand each step, you can run the complete version:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import
LinearRegression

# Step 1: Create DataFrame
data = {
    "Experience": [1, 2, 3, 4, 5, 6,
7, 8, 9, 10],
    "Salary": [25000, 28000, 32000,
35000, 40000,
               45000, 50000, 55000,
60000, 65000]
}

df = pd.DataFrame(data)

# Step 2: Define X and y
X = df[["Experience"]]
y = df["Salary"]

# Step 3: Create model
model = LinearRegression()

# Step 4: Train model
```

```
model.fit(X, y)

# Step 5: Print coefficient and
intercept
print("Slope:", model.coef_[0])
print("Intercept:", model.intercept_)

# Step 6: Print equation
print("Equation:")
print(f"Salary = {model.coef_[0]:.2f}
* Experience +
{model.intercept_:.2f}")

# Step 7: Predict salary for 7 years
experience
new_data = [[7]]
prediction = model.predict(new_data)

print("Predicted Salary for 7 years:",
prediction[0])

# Step 8: Plot
plt.scatter(X, y, label="Actual
Salary")
plt.plot(X, model.predict(X),
label="Regression Line")

plt.xlabel("Experience (Years)")
```

```
plt.ylabel("Salary")
plt.title("Linear Regression:
Experience vs Salary")

plt.legend()
plt.show()
```

The main concept to remember:

```
DataFrame
   ↓
X = Experience
y = Salary
   ↓
LinearRegression()
   ↓
model.fit(X, y)
   ↓
model.predict(X)
   ↓
Scatter Plot + Regression Line
```